🐍 Python Fundamentals
Values, Expressions & Statements

A value is any data stored in Python.

Types of Values:

- Numbers → 10, 3.14
- Boolean → True, False
- String → "Hello"

Example:

10 "Python" True

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Numbers in Python

1

Integer (int)

Whole numbers

5 -10 100

2

Float (decimal)

3.5 0.99

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Boolean (True / False)

Boolean has only two values:

True , False

Used in conditions:

```
5 > 2 # True
```

```
10 < 3 # False
```

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Strings

String = Text inside quotes

> "Hello" 'Python' "My name is Ali"

Strings are always inside single or double quotes

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Expressions

An expression is a combination of values and operators that gives a result.

Expression :	Result:
5 + 3	8
10 - 2	8
4 * 2	8

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Statements

A statement is an instruction Python executes.

Example:

```
x = 10
```

```
print(x)
```

x = 10 is a statement

print(x) is also a statement

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Operators

Operators perform operations.

+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Modulus
**	Power

Example:

```
2 ** 3 # 8
```

```
10 % 3 # 1
```

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Variables

A variable stores a value.

```
name = "Ali"
```

```
age = 20
```

Rules:

- Cannot start with number
- No spaces
- Use underscore _

Correct:

```
student_name = "Ahmed"
```

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Keywords

Keywords are reserved words in Python.

Examples:

- if
- else
- for
- while
- True
- False
- def

You cannot use keywords as variable names.

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